

# YIXUAN HE

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Assistant Professor at Arizona State University

## EDUCATION

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**University of Oxford, OU**

Oct. 2020 - Jun. 2024

*D.Phil. (PhD) student in Statistics*

*Supervisors: Mihai Cucuringu and Gesine Reinert*

- **Clarendon Scholar (fully funded)**, Balliol Jason Hu Scholarship
- Dissertation title: **Graph Neural Networks for Network Analysis**

**University of Edinburgh, UoE**

Sept. 2018 - Jun. 2020

*BSc (Hons) in Math and Stats*

*Supervisor: Timothy I. Cannings*

- GPA: **92.33%** Avg., **Rank # 1** in Math & Stats, **First Class**

**University of California, Berkeley**

Jan. 2018 - May 2018

*Semester Exchange*

- GPA: **4.0/4.0**

**South China University of Technology, SCUT**

Sept. 2016 - Jun. 2020

*BSc Mathematics and Applied Mathematics*

*Supervisor: Delu Zeng*

- Rank: **1/60** (Outstanding graduate)
- Overall GPA for 2016 - 18: **3.99/4.0**, Major GPA: **4.0/4.0**; transferred to UoE on 2+2 program
- Winning 2 years of **National Scholarship** granted by Ministry of Education of China
- **650/710** in CET6, **top 0.02 % of students** in SCUT

## RESEARCH EXPERIENCE

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***OU: Graph Neural Networks for Network Analysis***

Oct. 2020 - Jun. 2024

*Doctoral Research*

*Supervisors: Gesine Reinert, Mihai Cucuringu*

- Worked on signed, directed, and temporal GNNs, as well as GNNs for signal recovery.
- Developed open-source libraries: creator of PyG Signed Directed and contributor of PyG Temporal.
- Collaborated with Amazon as an applied scientist intern (Mentor: David Wipf).

***NetEase Games: Behavior-based Anti-cheating in FPS Games***

Jun. 2020 - Aug. 2020

*Summer Internship*

*Supervisor: Xin Wen*

- Extracted features from logs using multiprocessing and built multi-input single-output models with TensorFlow 2.2.

***UoE: AUROC vs AUPRC: A Comparison of Performance Measures in Classification Problems***

Sept. 2019 - Apr. 2020

*UoE Honors Project*

*Supervisor: Timothy I. Cannings*

- Collected data, compared measure scores, **analyzed measure properties** and data properties.

***UCLA CSST summer research/REU: Deforestation Brazil***

Jun. 2019 - Sept. 2019

*Funded by NGA and NSF*

*Supervisors: Christian Parkinson, Andrea L. Bertozzi, Stanley J. Osher*

- Conducted data mining, data analysis, MATLAB programming and model construction.
- Implemented e.g., **level set method**, numerical PDEs, optimal path searching, satellite image processing, Maximum Penalized Likelihood Estimation (MPLE), and time series simulation.

- **Delivered a talk** at IPAM’s “The Level Set Collective” seminar invited by Professor Stanley Osher.
- **Presented a poster** at Intelligence Community Academic Research Symposium (ICARS) 2019.

***SCUT: Robust Image Salient Object Perception Prior for Image Segmentation based on PDEs Related Methods***

May 2018 - May 2020

*Funded by National Natural Science Foundation of China (NSFC)*

*Supervisor: Delu Zeng*

- Studied level set methods and **neural networks** on **semantic image segmentation**.
- Worked on lung nodule segmentation using MEnet.
- Utilized Linux, Colab, Caffe, TensorFlow, PyTorch, and programming in Python.

***SCUT: Efficiency of Collaborative Innovation Knowledge Output in Professional Towns in Internet + Times***

May 2017 - May 2018

*Funded by MOE of PRC*

*Supervisors: Shifu Wang, Miaoxi Zhao, Shenquan Liu*

- **Cross-disciplinary:** Math and Urban Planning.
- Responsible: Data mining and analysis, paper writing.
- **Led** the project as part of the National Training Program of Innovation for Undergraduates.

## MAJOR AWARDS

|                        |                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------|
| <b>Grant</b>           | 2026 NVIDIA Academic Grant Program Award                                                    |
| <b>Grant and Honor</b> | 2026 Jetstream2 NAIRR AI Fellows Program                                                    |
| <b>Grant</b>           | 2024 G-Research June Grant (mainly for attending ICML 2024)                                 |
| <b>Grant</b>           | 2024 ICML Official Financial Aid                                                            |
| <b>Grant</b>           | 2024 ICLR Official Financial Assistance                                                     |
| <b>Award</b>           | 2023/2024 Academic Year Oxford University Blues Award for Sports Excellence                 |
| <b>Grant</b>           | 2024 Balliol College Support Grant                                                          |
| <b>Grant</b>           | 2023-24 Balliol College Floreat Grant for Sports                                            |
| <b>Grant</b>           | 2022-23 Balliol College Floreat Grant for Sports                                            |
| <b>Grant</b>           | 2022 ICML Official Participant Award                                                        |
| <b>Grant</b>           | 2022 Simons Institute Summer School Fund                                                    |
| <b>Grant</b>           | 2022 Balliol’s fund on IMA Conf. on The Mathematical Challenges of Big Data reg.            |
| <b>Award</b>           | 2022 SIAM Student Travel Award for SDM’22                                                   |
| <b>Award</b>           | 2022 SIGIR Student Registration Award for WSDM’22                                           |
| <b>Grant</b>           | 2021 Balliol’s fund on Complex Networks 2021 reg.                                           |
| <b>Grant</b>           | 2021 Balliol’s fund on NeurIPS 2021 reg.                                                    |
| <b>Grant</b>           | 2021 Balliol’s fund on ACM membership and CIKM’21 registration                              |
| <b>Prize</b>           | 2020 Napier Medal and Gangadhar Balwant Gadgil Prize ( <b>top</b> Math student at UoE)      |
| <b>Prize</b>           | 2020 Lawley Memorial Prize ( <b>top</b> student in Math and Stats at UoE)                   |
| <b>Scholarship</b>     | 2020 <b>Clarendon Scholarship</b> and Balliol Jason Hu scholarship at Oxford                |
| <b>Honor</b>           | 2020 Outstanding Graduate at SCUT Math Dept.                                                |
| <b>Prize</b>           | 2019 Arthur Erdelyi Prize (For top three performing Math students at UoE)                   |
| <b>Prize</b>           | 2019 James Ward Memorial Prize ( <b>top</b> student in Math and Stats at UoE)               |
| <b>Second Prize</b>    | 2018 ICM/MCM International <b>Mathematical Modeling</b> Competition                         |
| <b>Grand Prize</b>     | 2018 Math Competition of SCUT (Math Group) ( <b>Rank # 1</b> )                              |
| <b>Honor</b>           | 2017-2018 one of the 10 <b>Role Models to Merit Students</b> at SCUT                        |
| <b>Prize</b>           | 2017 and 2018 <b>National Scholarship</b>                                                   |
| <b>Second Prize</b>    | 2017 “9 <sup>th</sup> National College Student <b>Math Competition</b> (Math)” in Guangdong |
| <b>Second Prize</b>    | 2017 National College Student Mathematical Modeling Competition (Guangdong)                 |

## PUBLICATIONS

\* Means equal contribution. Non-arXiv publications are peer-reviewed. “Talk” means additional conference presentations.

- Sandel, A., **He, Y.**, ..., & Mitani, J. Lethal Conflict Following Group Fission in Wild Chimpanzees. (**Science**, 2026)
- Liang, P., Yang, C., **He, Y.**, Pan, R., & Chen, Y. Topology-Guided Graph Pre-training and Prompt Learning on Directed Graphs. (**TMLR**, 2026)
- Yilmaz, N., Patel, M., Kusumba, A., **He, Y.**, & Yang, Y. MentalBlackboard: Evaluating Spatial Visualization via Mathematical Transformations. (arXiv 2026)
- Abolhasani, M., Ba, Y., **He, Y.**, & Pan, R. Beyond Predefined Schemas: TRACE-KG for Context-Enriched Knowledge Graphs from Unstructured Documents. (arXiv 2026)
- Ockerman, S., Gueroudji, A., Mallick, T., **He, Y.**, Pouchard, L., Ross, R., & Venkataraman, S. PGT-I: Scaling Spatiotemporal GNNs with Memory-Efficient Distributed Training. (**SC25**)
- **He, Y.**, Sandel A., Wipf D., Cucuringu M., Mitani J. & Reinert G. Learning to Fuse Temporal Proximity Networks: A Case Study in Chimpanzee Social Interactions. (arXiv 2025)
- Aminian, G.\*, **He, Y.\***, Reinert, G., Szpruch, L., Cohen, SN. Generalization Error of Graph Neural Networks in the Mean-field Regime. (**ICML 2024**)
- **He, Y.**, Reinert, G., Wipf, D., & Cucuringu, M. Robust Angular Synchronization via Directed Graph Neural Networks. (**ICLR 2024**)
- Steach, H., Viswanath, S., **He, Y.**, Zhang, X., Ivanova, N., Hirn, M., Perlmutter, M., & Krishnaswamy, S. Inferring Metabolic States from Single Cell Transcriptomic Data via Geometric Deep Learning. (**RECOMB 2024**)
- **He, Y.**, Zhang, X., Huang, J., Rozemberczki, B., Cucuringu, M., & Reinert, G. PyTorch Geometric Signed Directed: A Software Package on Graph Neural Networks for Signed and Directed Graphs. (**LoG 2023, Open-Source Software with 100+ stars**)
- Geng, H., Chen, C., **He, Y.**, Zeng, G., Han, Z., Chai, H., Yan, J. Pyramid Graph Neural Network: A Graph Sampling and Filtering Approach for Multi-scale Disentangled Representations. (**KDD 2023, Talk**)
- Wu, Q., Yang, C., Zhao, W., **He, Y.**, Wipf, D., & Yan, J. DIFFormer: Scalable (Graph) Transformers Induced by Energy Constrained Diffusion. (**ICLR 2023, Spotlight Talk**)
- **He, Y.**, Reinert, G., & Cucuringu, M. DIGRAC: Digraph Clustering Based on Flow Imbalance. (**LoG 2022**)
- **He, Y.**, Perlmutter, M., Reinert, G., & Cucuringu, M. MSGNN: A Spectral Graph Neural Network Based on a Novel Magnetic Signed Laplacian. (**LoG 2022**)
- Wang, X., Chen, S., **He, Y.**, Wang, M., Gan, Q., & Yan, J. CEP3: Community Event Prediction with Neural Point Process on Graph. (**LoG 2022**)
- **He, Y.**, Gan, Q., Wipf, D., Reinert, G. D., Yan, J., & Cucuringu, M. GNNRank: Learning Global Rankings from Pairwise Comparisons via Directed Graph Neural Networks. (**ICML 2022, Talk**)
- **He, Y.**, Reinert, G., Wang, S., & Cucuringu, M. SSSNET: Semi-Supervised Signed Network Clustering. (**SDM 2022, Talk**)
- **He, Y.** GNNs for Node Clustering in Signed and Directed Networks. (**WSDM-DC 2022, Talk**)
- Zhang, X., **He, Y.**, Brugnone, N., Perlmutter, M., & Hirn, M. MagNet: A Neural Network for Directed Graphs. (**NeurIPS 2021**)
- Rozemberczki, B., Scherer, P., **He, Y.**, Panagopoulos, G., Riedel, A., Astefanoaei, M., ... & Sarkar, R. PyTorch Geometric Temporal: Spatiotemporal Signal Processing with Neural Machine Learning Models. (**CIKM 2021 best paper award, Open-Source Software with 2.9k stars, 400+ forks**)
- **He, Y.**, Hu, T., & Zeng, D. Scan-flood Fill (SCAFF): An Efficient Automatic Precise Region Filling Algorithm for Complicated Regions. (**CVPR 2019 workshop**)
- Zeng, D., **He, Y.**, Liu, L., Chen, Z., Huang, J., Chen, J., & Paisley, J. Ro-SOS: Metric Expression Network (MEnet) for Robust Salient Object Segmentation. (arXiv 2018)

## OTHER SELECTED ORAL PRESENTATIONS

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- Department of Epidemiology and Biostatistics Program Seminar Series, University of Arizona, 2025
- Meta Research Seminar, Meta, 2025
- Learning on Graphs and Geometry seminar, OU, 2025
- UCSC Statistics Seminar, University of California, Santa Cruz, 2025.
- Cornell CS Theory Tea, Cornell University, 2025.
- Research Innovation in the Mathematical Sciences Seminar, ASU, 2024.
- CAM / DoMSS Seminar, ASU, 2024.
- Statistics Seminar, ASU, 2024.
- CNRS GdR IASIS Graph Learning Day, CNAM Paris, 2024.
- Statistical learning on LARge scale GRaphs (LARGR), Inria Lille - Nord Europe, 2023.
- 4th IMA Conference on The Mathematical Challenges of Big Data, OU, 2022.
- The 10th Int. Conf. on Complex Networks and their Applications, Spain (virtual), 2021.

## TEACHING EXPERIENCE

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### ASU's Assistant Professor

Sept. 2024 -

- Grad course: Statistics for Bio Data Sci II (F24)
- Undergrad courses: Regression and Time Series Analyses (F25), Stat Analysis for Researchers (SP25, SP26), Nonparametric Statistics (SP25), Statistical Computing (SP26)

### OU's Teaching Assistant

Dec. 2020 - Dec. 2022

- Courses to grade and discuss a problem in the problem classes: Applied Statistics (F22), Advanced Topics in Statistical Machine Learning (SP22), Computational Statistics (SP21)
- Course to demonstrate and answer questions: Statistics and Data management (for DTC in Dept. of Zoology, F20)

### UoE's Digital Skills Trainer - Python

Dec. 2019 - May 2020

- **Delivered Python training courses**, reviewed and improved existing Python training.
- Recorded lectures: Introduction to Python and Python for Data Science.

### Teaching Support in a Primary School in Longsheng Rural Village

July 2017

- Devised and delivered lectures in singing, dancing, craftsmanship, and storytelling.
- **Leader** in hosting the closing ceremony.
- **Visited families** to collect information and offer suggestions.

## ACADEMIC SERVICES

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### Area Chair (AC)/Reviewer

- Conferences: GFM at ICML 2026 (AC), ICML 2026, ICLR 2026, AISTATS 2026, AAAI 2026, NeurIPS 2025, ICML 2025, AISTATS 2025, ICLR 2025, AAAI 2025, NeurIPS 2024, ICML 2024, ICLR 2024, AAAI 2024, NeurIPS 2023, ICML 2023, WWW 2023, AAAI 2023, NeurIPS 2022, SIGKDD 2022, ICML 2022, WSDM 2022
- Journals: IEEE Transactions on Signal and Information Processing over Networks, IEEE Transactions on Neural Networks and Learning Systems, IEEE Transactions on Knowledge and Data Engineering, Journal of Complex Networks, Journal of Quality Technology, Statistics and Computing

**(Co-)Organizer**

- Learning on Graphs Conference 2026 (program chair)
- ASU Machine Learning Day 2026, ASU
- Learning on Graphs Conference 2025, ASU (local host)
- ASU Machine Learning Day 2025, ASU
- AI Seminar at ASU, ASU
- Learning on Graphs Local Meetup at Arizona 2024, ASU
- LoG<sup>2</sup> at Oxford: the Learning on Graphs and Geometry seminar series, OU

**WORK EXPERIENCE****Assistant Professor at ASU**

Sept. 2024 -

- Teach Statistics at School of Mathematical and Natural Sciences.
- Conduct research and mentor students.
- Organize and participate in research activities.

**Applied Scientist Intern at Amazon Web Services**

Nov. 2021 - Aug. 2024

- Mentor: David Wipf.
- Conducted research on graph neural networks.

**Game Development Engineer at NetEase Games**

Jun. 2020 - Aug. 2020

- Project: Behavior-based Anti-cheating in FPS Games (Supervisor: Xin Wen)
- Selected as **Outstanding Intern** to present my work.

**Student Assistant for School of Mathematics at SCUT**

Sept. 2016 - Aug. 2017

- Assisted in organizing materials and releasing notifications.
- Outstanding individual in work-study of South China University of Technology.

**CONFERENCES/SUMMER SCHOOLS**

|                                                                           |                          |            |
|---------------------------------------------------------------------------|--------------------------|------------|
| 2026 AAAS Annual Meeting – Science at Scale                               | Volunteer                | Feb. 2026  |
| 4th Learning on Graphs (LoG) Conference                                   | Organizer and Local Host | Dec. 2025  |
| 41st International Conference on Machine Learning ( <b>ICML</b> )         | Author                   | Jul. 2024  |
| CNRS GdR IASIS Graph Learning Day                                         | Speaker                  | Jun. 2024  |
| 12th International Conference on Learning Representations ( <b>ICLR</b> ) | Author                   | May 2024   |
| 2nd Learning on Graphs (LoG) Conference                                   | Author                   | Nov. 2023  |
| Workshop: statistical learning on LARge scale GRaphs (LARGR)              | Presenter                | Mar. 2023  |
| 1st Learning on Graphs (LoG) Conference                                   | Author                   | Dec. 2022  |
| 4th IMA Conference on The Mathematical Challenges of Big Data             | Speaker                  | Sept. 2022 |
| 39th International Conference on Machine Learning ( <b>ICML</b> )         | Author                   | Jul. 2022  |
| SIAM International Conference on Data Mining ( <b>SDM</b> )               | Author                   | Apr. 2022  |
| 15th ACM Int. Conf. on Web Search and Data Mining ( <b>WSDM</b> )         | Doctoral Consortium      | Feb.2022   |
| 35th Conf. on Neural Information Processing Systems ( <b>NeurIPS</b> )    | Author                   | Dec.2021   |
| The 10th Int. Conf. on Complex Networks and their Applications            | Author                   | Nov.2021   |
| Int. Conf. on Information and Knowledge Management ( <b>CIKM</b> )        | Author                   | Nov.2021   |
| Intelligence Community Academic Research Symposium ( <b>ICARS</b> )       | Poster Presenter         | Sept. 2019 |
| Computer Vision and Pattern Recognition ( <b>CVPR</b> )                   | Author                   | Jun. 2019  |

## COMPUTER SKILLS

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Programming: **Python (Advanced)**, C++ (Intermediate), VB (Basic), and Pascal (Basic)  
 Statistical Analysis: **R (Advanced)**, SQL (Intermediate), and SPSS (Intermediate)  
 Modelling: **Matlab (Advanced)**, Maple (Intermediate), and Xpress (Intermediate)  
 Paper writing: **Latex (Advanced)**

## POSITIONS OF RESPONSIBILITY

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**President of Oxford University Table Tennis Club (OUTTC)** May 2023 - May 2024

- **Organized** club activities, promoted the club, organized training, managed booking, and coordinated facilities.
- Participated in table tennis **competitions** as a squad member.
- **Initialized** the use of an Oxford mailing list to send notifications (600+ subscribers) and **managed** a club Facebook group (1.2k members).

**Director of Dept. of Visiting Scholars in Oxford Chinese Students and Scholars Association (OXCSSA)** July 2021 - July 2022

- **Co-organized** 2021 welcoming event for new students to OU.
- **Organized** events for visiting scholars: e.g., visiting scholars' tea party, hiking, formal, and punting.
- Collaborated with other departments to hold virtual and in-person talks.

**Head of Class** Sept. 2013 - Jun. 2018

- Conveyed notifications, organized activities, and led by example.
- 2017 and 2018 South China University of Technology **Outstanding Student Leader**.
- 2016-2017 **outstanding class** of SCUT and 2017-2018 top-16 classes of SCUT.

**University Student Representative** May 2017 - May 2018

- Actively listened to students' opinions and participated in raising proposals.
- Outstanding Student Representative of South China University of Technology in 2017.

**Member of Public Relationship Dept., Math Student Union** Sept. 2016 - Oct. 2017

- **Led** the hosting of a social party in School of Mathematics.
- 2017 **Star** in Mathematics Student Union.